Fnu Tushar

Malvern, PA | +1 (484) 716-7840 | fnu.tushar@outlook.com | linkedin.com/in/fnu-tushar-945648131 | github.com/tus2014ar

Professional Summary

Data engineer with 5+ years building data pipelines and data quality checks (SQL, Python, Databricks). MS in Data Analytics, Penn State (Dec 2026); seeking full-time Data Engineer roles.

Technical Skills

Programming: Python (Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, TensorFlow, Keras), SQL, R, Java.
Data Engineering: Databricks, Unity Catalog, Delta Lake, PySpark, ETL/ELT, Dimensional Modeling, Medallion Architecture, Data Warehousing, Oracle Database, SQLite.
Cloud & DevOps: Azure, CI/CD, GitHub, GitHub Actions, Jira.
Analytics & AI: Tableau, Excel, EDA, NLP, SBERT, LLM Integration (Llama 3), Claude API.
AI Productivity Tools: ChatGPT, Julius AI, Cursor IDE, Claude Code, GitHub MCP, Claude CLI, Claude Skills.

Work Experience

Barton Malow May 2026 – Aug 2026
Intern - Data Science *Michigan, United States*

- **Engineered a Unity Catalog billing pipeline:** unowned Databricks spend cut 85.6% to 54%, $146K under governance.
- **Attributed 8,000+ Claude queries** (20+ users) on shared Databricks compute, exposing AI's warehouse-spend share growing 5.5x (1.5% to 8.3%) in 5 months.
- **Built a 211-check data quality framework** on 23 tables: ~84% of active records had issues; 39% completeness bug found.

Pennsylvania State University Sep 2025 – May 2026
Research Assistant - Data Science *Pennsylvania, United States*

- **Built a local-first audit AI-framework**: 5 CMS/NIST rules, Llama 3 explanations, append-only SHA-256 SQLite log.
- **Automated RA candidate screening** with a three-stage TF-IDF to SBERT pipeline matching 120 resumes to 2,277 projects at 96% accuracy.

Oracle Corporation Feb 2022 – Dec 2024
Staff Data Consultant *Bangalore, India*

- **Validated billing data pipelines** for 600K+ telecom subscribers using SQL quality checks, anomaly rules and reconciliation: 98%+ accuracy within SLAs.
- **Prepared and automated churn and usage KPI dashboards** for weekly leadership reviews: 99% on-time delivery.
- **Led onsite testing** for a 15-member team through a 1-month go-live; presented a declining post-launch bug trend.

Accenture Solutions Pvt Ltd Jun 2019 – Feb 2022
Data Consultant *Pune, India*

- **Resolved 50+ recurring data quality issues** in production billing database for 4.5M customers, improving billing accuracy.
- **Automated batch bill-run workflows** in SQL: 1 day to 2 hours (12x faster), manual effort down 92%.
- **Designed a SQL validation framework** (row-level checks, reconciliation, exception flags): discrepancies down 60%, month-end close 10% faster.

Education

Pennsylvania State University Pennsylvania, United States
Master of Data Analytics, GPA: 3.96/4 *Aug 2025 – Dec 2026*
Statistical Analysis, Data-Driven Decision Making, Predictive Analytics, Deep Learning, Natural Language Processing, AI Ethics.

KIIT University Odisha, India
Bachelor of Technology in Electronics and Instrumentation Engineering, GPA: 3.3/4 *May 2015 – May 2019*
Control Systems, Digital Electronics, OOP, Data Structures & Algorithms, Artificial Intelligence.

Projects

Osteoarthritis Market Share Forecasting Pipeline GitHub
Re-runnable data platform for an OA injectable brand manager: IQVIA visit data and 171 public datasets in one audited, quality-checked warehouse that feeds forecasts and an analytics site.

- **Engineered a re-runnable Bronze/Silver/Gold pipeline:** 149K+ IQVIA rows to a star schema, 11 data quality checks.
- **Modeled a 5-dimension, 10-fact schema** from 171 public files (~24 GB), with 38 reconciliation checks passing.
- **Implemented a serving layer on warehouse:** AI answers via 11 read-only tools on a sourced 311-node, 827-edge graph.

Worldwide Sentiment Analysis of LLM Tweets GitHub
Data pipeline behind an LLM-sentiment analysis of 305K tweets: profiles and normalizes noisy locations, then feeds regional sentiment scoring and forecasting for AI policy teams.

- **Profiled 305K tweets** (36 languages): 33,739 unique location strings, 25% missing, 2,162 duplicate texts.
- **Built a location-normalization pipeline** (aliases, bio inference, geocoding): Other/Unknown 72.8% to 43.9%.
- **Fed cleaned data to VADER scoring and a SARIMAX forecast** (weekly seasonality) behind an AI-policy brief.

Publications

First-authored "Autonomous Multi-Agent Governance: AI Sentinel Framework for Mitigating Psychological Restraints in AI-Driven Fall Management," Springer CCIS vol. 3126, SEET 2026, with industry partner MyRaina Inc.